

# A Rights-Based Framework for Ethical AI Training Data Governance

## A Three-Tier Licensing Model Bridging Innovation and Creator Rights

**Margot MacKechnie** Whitman School of Management, Syracuse University B.S. Entrepreneurship & Emerging Enterprises | Minor in Information Technology, Design & Startups February 2026

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### Executive Summary

The rapid advancement of artificial intelligence has created urgent questions about intellectual property rights, creator compensation, and ethical data use. Current AI training practices frequently utilize copyrighted content without explicit permission or compensation, generating significant legal challenges and raising fundamental ethical concerns about the relationship between human creativity and machine learning.

This white paper proposes a comprehensive three-tier framework for ethical AI training data governance, modeled on two proven precedents: the European Union's data privacy regulatory architecture (particularly GDPR) and the music industry's collective rights management systems (ASCAP, BMI, mechanical licensing). The goal is to give creators genuine agency over their intellectual property while preserving the conditions that enable AI innovation to continue.

### The Framework at a Glance:

- **Tier 1 - Active Licensing:** Creators opt in to license their work for AI training and receive compensation proportional to usage, through collective management organizations modeled on music royalty infrastructure.
  - **Tier 2 - Explicit Opt-Out:** Creators can formally exclude their work from training datasets, backed by legal enforcement mechanisms and technical protocols modeled on GDPR's Right to Object and Right to Erasure.
  - **Tier 3 - Attribution and Transparency:** A baseline default for all other works, requiring AI companies to maintain comprehensive training data documentation and give creators the ability to verify usage, grounded in EU moral rights traditions and GDPR transparency obligations.
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## 1. Background and Motivation

### 1.1 The Problem

Modern large-scale AI models are trained on vast datasets scraped from the internet, digitized books, creative archives, and other sources containing the work of millions of human creators. In most cases, these creators

were not asked for permission, are not notified that their work was used, and receive no compensation.

This practice has generated significant controversy on several fronts:

**Legal:** Authors, visual artists, musicians, and news publishers have filed suit against AI developers in the United States and Europe, alleging copyright infringement. The outcomes of these cases will shape the regulatory landscape for years to come, but the legal framework is still unsettled.

**Ethical:** Even where AI companies may have technical legal defenses (such as fair use arguments), many creators argue that using their life's work to build commercially valuable systems without consent or compensation is fundamentally unjust, regardless of legality.

**Economic:** As AI-generated content proliferates, it increasingly competes directly with the human creators whose work trained the systems. Writers, illustrators, coders, and musicians face market displacement by tools that were built on their own output.

**Regulatory:** The EU AI Act (2024) and existing GDPR obligations create a complex compliance environment for AI developers operating in or selling to European markets. The lack of a clear, standardized framework for training data governance creates compliance risk and legal uncertainty on all sides.

## 1.2 The Gap This Framework Addresses

Existing proposals tend to fall into two camps: those that defend current AI training practices under broad fair use or transformative use theories, and those that call for strict opt-in requirements that critics argue would make large-scale AI training practically impossible. This framework proposes a middle path, one that takes creator rights seriously while designing practical, scalable mechanisms for compliance.

The music industry offers a compelling precedent. For decades, it has navigated the tension between widespread use of creative works and fair compensation through collective licensing organizations, compulsory licensing regimes, and digital rights metadata standards. These systems are imperfect, but they demonstrate that high-volume use of copyrighted work and meaningful creator compensation are not mutually exclusive.

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## 2. Regulatory Foundation

This framework draws on several existing legal and regulatory instruments:

**GDPR (General Data Protection Regulation, EU 2016/679):** The world's most comprehensive data rights framework establishes principles including transparency, consent, right of access, right to erasure, data minimization, and accountability that translate directly into a training data governance context, even where personal data is not involved.

**EU AI Act (2024):** Requires providers of general-purpose AI models to publish summaries of training data used for pretraining and fine-tuning, and to comply with EU copyright law. Providers must also put in place policies to comply with opt-out requests from rights holders.

**EU Copyright Directive (2019/790), Article 4:** Permits text and data mining for research purposes, but allows rights holders to reserve their rights for commercial applications, establishing a legal basis for opt-out mechanisms.

**EU Moral Rights Tradition:** Unlike US copyright law, EU member states protect the moral rights of authors, including the right of attribution and the right to object to derogatory treatment of their work, which survive even the transfer of economic rights.

**Music Industry Licensing Models:** Collective management organizations (CMOs) such as ASCAP, BMI, SESAC (US), PRS for Music (UK), and STIM (Sweden) demonstrate viable infrastructure for high-volume licensing, proportional compensation, and rights tracking at scale.

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### 3. The Three-Tier Framework

#### Tier 1: Active Licensing (Opt-In with Compensation)

**Core Principle:** Creators who wish to actively participate in AI training markets can register their works and receive proportional compensation, maintaining full ownership and auditing rights throughout.

#### Mechanism:

Creators register works in a centralized AI Training Data Registry modeled on existing CMO infrastructure, with metadata specifying content type, licensing terms, and compensation requirements. AI companies access registered works through blanket licensing agreements or micro-licensing arrangements, paying into collective pools distributed to rights holders on a proportional basis.

#### Compensation Models:

- Micro-licensing: Pay-per-use based on frequency and weighting in training
- Blanket licensing: Annual fees for access to registered creator pools, similar to streaming blanket licenses
- Tiered compensation: Higher rates for rare, specialist, or high-quality content; standard rates for common content types

#### Creator Rights Under Tier 1:

- Moral rights retained (attribution, integrity)
- Right to audit AI company's usage records
- Right to withdraw consent with reasonable notice
- Right to receive annual usage and earnings reports

#### Transparency Obligations on AI Companies:

- Disclose which registered works were included in training
  - Report approximate influence and weighting of each work
  - Publish commercial applications of models trained on licensed data
  - Maintain records subject to third-party audit
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## **Tier 2: Explicit Opt-Out (Protected Rights)**

**Core Principle:** Creators who do not wish their work used in AI training can formally exclude it, with legal enforcement mechanisms and technical compliance requirements backed by meaningful penalties, modeled on GDPR's Right to Object and Right to Erasure.

### **Mechanism:**

Creators designate works as opt-out through a standardized technical protocol (analogous to robots.txt but legally binding for AI training) and/or through formal registration with the AI Training Data Registry. AI companies are required to maintain up-to-date opt-out lists and implement technical controls to exclude designated works from training pipelines.

### **Enforcement:**

- Violations treated as analogous to GDPR data processing violations
- Mandatory removal of works from training datasets upon verified opt-out
- Fines calculated on a per-work, per-model basis
- Potential model retraining or deployment suspension for systematic non-compliance
- Private right of action for affected creators

### **Technical Requirements on AI Companies:**

- Content fingerprinting and hash-based exclusion systems
- Regular opt-out list updates (minimum quarterly)
- Pre-training compliance audits by independent third parties
- Public reporting on opt-out request processing

**Important Distinction from Tier 3:** Opt-out is not a default. It requires an affirmative act by the creator. This preserves the practical feasibility of AI training while giving creators a meaningful, legally enforceable right to withdraw.

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### **Tier 3: Attribution and Transparency (Baseline Default)**

**Core Principle:** For all works not covered by Tier 1 or Tier 2, a baseline standard applies: AI companies must maintain comprehensive records of training data, creators have the right to verify whether their work was used, and attribution metadata must be preserved and accessible.

#### **Mechanism:**

This tier applies automatically. It requires no action from creators but imposes affirmative obligations on AI developers. It is grounded in EU moral rights traditions and GDPR transparency principles (Articles 13-14), adapted to the training data context.

#### **AI Company Obligations:**

##### *Training Documentation:*

- Maintain comprehensive logs of all works used in training, with creator attribution where determinable
- Record approximate influence and weight of works in the model
- Preserve records for a minimum of 10 years following model deployment

##### *Public Attribution Database:*

- Maintain a searchable registry enabling creators to query whether specific works were included in training
- Provide creator-facing APIs for programmatic verification

##### *Model Cards:*

- Publish standardized documentation for each deployed model listing major training data sources, composition statistics, and known gaps

##### *Generated Content Metadata:*

- Where technically feasible, embed provenance metadata in AI-generated outputs identifying influential training sources

#### **Creator Rights Under Tier 3:**

- Right to be informed upon request whether specific works were used
- Right to correct misattributions in training records
- Right to upgrade to Tier 1 (seeking compensation) or Tier 2 (opting out) at any time
- Right to contest outputs that too closely replicate original works



**Compensation:** Tier 3 does not guarantee individual compensation, but a collective levy system modeled on blank media levies used in several EU countries could fund a general pool distributed to creative communities proportional to the volume of their work used in training.

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## **4. Governance and Enforcement Architecture**

### **4.1 Regulatory Body**

A dedicated AI Training Data Authority (ATDA) should be established at the EU level (with national equivalents for non-EU jurisdictions), with the following mandate:

- Maintain the AI Training Data Registry across all three tiers
- Issue and update technical standards for opt-out protocols and attribution metadata
- Receive and investigate complaints from creators and AI companies
- Conduct periodic audits of major AI developers' compliance
- Publish annual transparency reports on the state of AI training data governance

### **4.2 Collective Management Organizations**

Existing CMOs should be empowered to negotiate blanket licensing agreements with AI companies on behalf of their members (Tier 1), analogous to their role in music performance rights. New CMOs may be needed for content categories such as visual art and journalism that lack established collective management infrastructure.

### **4.3 Dispute Resolution**

A streamlined dispute resolution process should be established for:

- Creator claims that their work was used without authorization
  - AI company challenges to opt-out designations
  - Attribution accuracy disputes
  - Compensation calculation disagreements
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## **5. Scope Clarification: Open Licensing, Public Domain, and Open Source Code**

A complete framework must address the significant categories of creative and technical work that already exist under permissive or open licensing terms. These works occupy a different position than proprietary, all-rights-reserved content and require distinct treatment.

## 5.1 Creative Commons Licensed Works

Creative Commons (CC) licenses are copyright licenses. They operate within the copyright system and grant specific permissions in advance. Their interaction with AI training is genuinely complex and has been the subject of ongoing analysis by Creative Commons itself, culminating in formal guidance published in 2025.

The core challenge is that in jurisdictions where AI training is covered by a copyright exception such as the EU's text and data mining (TDM) exception under Article 4 of the Copyright Directive, or the US fair use doctrine, the CC license terms may not even be triggered, since the license only applies where copyright permission is required. This means that a more restrictive CC license (such as CC BY-NC) may not effectively prevent commercial AI training in jurisdictions with broad TDM exceptions.

Nonetheless, this framework takes the position that AI developers should treat CC license conditions as meaningful signals of creator intent and comply with them in good faith, regardless of whether strict legal obligation exists. Specifically:

**CC BY (Attribution):** AI companies training on CC BY works must include attribution to the source dataset at minimum, and where technically feasible, should attribute at the individual work level. This aligns directly with the Tier 3 attribution obligations in this framework.

**CC BY-NC (NonCommercial):** Works designated NonCommercial should not be used in training datasets for commercially deployed AI models without explicit Tier 1 licensing. This framework treats the NC designation as equivalent to a Tier 2 opt-out signal for commercial AI applications.

**CC BY-ND (NoDerivatives):** The NoDerivatives condition applies where AI outputs constitute an adaptation of the original work. Where AI models produce outputs substantially similar to ND-licensed training works (a form of memorization), this raises compliance concerns regardless of jurisdiction. AI developers should implement memorization-mitigation techniques for ND-licensed training data.

**CC BY-SA (ShareAlike):** The interaction between ShareAlike conditions and AI training is particularly contested. Research by the Institute for Information Law at the University of Amsterdam (commissioned by Open Future, 2024) found that ShareAlike obligations are largely ineffective in current AI training workflows because most SA licenses do not apply where TDM exceptions exist. This framework supports Creative Commons' own initiative to develop a separate preference signals framework outside copyright law, a social contract mechanism that would allow rights holders to express AI training preferences independently of their CC license choice.

**CC0 (Public Domain Dedication):** Works released under CC0, where the creator has waived all rights to the fullest extent permitted by law, fall outside all three tiers of this framework. They may be freely used for AI training without restriction or obligation.

## 5.2 Public Domain Works

Works in the public domain, whether because copyright has expired, because they were created by government agencies in jurisdictions that do not recognize copyright in government works, or because rights holders have

voluntarily dedicated them, are similarly outside the scope of this framework. No licensing, compensation, or attribution obligations apply to public domain works under this framework, though AI developers are encouraged to document their use of public domain sources as a matter of good record-keeping.

One important nuance: digitized versions of public domain works may attract new rights in some jurisdictions (particularly through the EU's database right). This framework does not resolve that question, but notes that claims of new rights over digitized public domain content should be treated skeptically and are subject to ongoing legal contestation.

### **5.3 Open Source Software Code**

Open source code presents a distinct set of considerations that sit somewhat outside the scope of this framework's primary focus on creative works but warrant brief treatment, given the significant volume of open source code used to train AI coding assistants.

Open source licenses including the MIT License, Apache License 2.0, GNU GPL, and others impose specific conditions on the use and distribution of software. The application of these conditions to AI training is actively contested. Key questions include whether training an AI model on GPL-licensed code creates a derivative work that must be released under GPL (the copyleft question), and whether AI-generated code that reflects patterns learned from GPL-licensed training data triggers copyleft obligations.

This framework takes the position that:

- Permissive open source licenses (MIT, Apache 2.0, BSD) should be treated analogously to CC BY: attribution obligations apply but no restrictions on commercial use.
- Copyleft licenses (GPL, AGPL) raise unresolved questions that this framework does not definitively resolve, but AI companies training on copyleft-licensed code should proactively engage with the open source community and contribute to emerging norms in this space.
- The open source community's preference signals, including explicit statements by projects that they do or do not consent to AI training use, should be respected as Tier 2 opt-out signals, even absent a formal opt-out mechanism.

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## **6. Equity and Access: Small Creators vs. Large Publishers**

One of the most significant risks in any AI training data governance framework is that it inadvertently replicates or worsens the existing power imbalances between large, well-resourced content owners and the vast majority of creators who work independently or at small scale.

### **6.1 The Structural Problem**

The current AI licensing landscape illustrates this risk clearly. As of 2025, AI companies have struck major bilateral licensing deals primarily with large media corporations and academic publishers: the Associated Press,



News Corp, Springer Nature (\$23 million), Wiley (\$23 million), and Reddit (\$60 million annually from Google). These organizations have the legal resources, negotiating leverage, and institutional infrastructure to identify, enter, and enforce licensing agreements.

Individual creators, including independent writers, photographers, illustrators, musicians, bloggers, and freelance journalists, are effectively invisible in this market. As publishing industry analyst Jane Friedman has noted, AI companies are "primarily pursuing deals with large corporations, not individuals or small players." A 2025 study commissioned by CISAC (the International Confederation of Societies of Authors and Composers, representing over 5 million creators) found that music and audiovisual creators stand to lose 24% and 21% of their revenues respectively by 2028 due to AI substitution, yet these are precisely the creators least equipped to negotiate individual licensing terms.

Research published at the 2025 ACM CHI Conference found that creative workers in smaller organizations and freelance settings lack both the AI governance support and the collective bargaining infrastructure that larger employers sometimes provide. Freelancers in particular face a structural disadvantage: in most jurisdictions, they cannot legally unionize in the traditional sense, leaving them without the collective voice needed to negotiate meaningful terms with AI companies.

Without deliberate design to address this asymmetry, any governance framework risks becoming a system that works well for Springer Nature and poorly for the independent academic blogger whose work was scraped alongside it.

## **6.2 Mechanisms for Structural Equity**

This framework proposes the following specific mechanisms to address the small creator equity problem:

### **Simplified Registration Pathways**

The AI Training Data Registry must offer a frictionless registration process accessible to individual creators without legal or technical expertise. This means:

- One-click opt-out signals embedded in standard website publishing tools (WordPress, Substack, Squarespace, etc.), analogous to Cloudflare's AI bot blocking tool launched in 2025
- Automatic registration of works published on participating platforms, so creators do not need to individually register each piece
- Free registration for individual creators and small publishers; costs for commercial-scale registration borne by AI companies through licensing fees

### **Collective Representation**

Individual creators should have access to collective management organizations (CMOs) that can negotiate on their behalf, in the same way that ASCAP and BMI negotiate performance rights for songwriters who could never individually license their music to every radio station. This framework calls for:

- Expansion of existing CMO mandates to cover AI training licensing (building on the Copyright Clearance Center's 2024 move to include AI training rights in its licensing offerings)
- Establishment of new CMOs for content categories such as visual art and journalism that lack adequate collective management infrastructure
- Anti-fragmentation rules requiring AI companies to engage with collective licensing bodies rather than only pursuing individual deals with large publishers, which risks establishing a two-tier system of licensed and unlicensed creators

### **Proportional Revenue Distribution**

Compensation distributed through Tier 1 collective pools must include protections against the revenue concentrating entirely at the top of the creator hierarchy. This means:

- Minimum per-creator floor payments regardless of individual work volume
- Graduated distribution formulas that weight smaller creators' contributions proportionally rather than purely by volume
- Transparent reporting on how distributions are calculated, enforceable by the ATDA

### **Tiered Compliance Costs**

AI companies' obligations under this framework should scale with their commercial scale. A large foundation model provider generating billions in revenue should face full compliance requirements. A nonprofit research institution or an early-stage startup should face simplified, lower-cost pathways, with the caveat that commercial deployment triggers full obligations regardless of company size or age.

### **Platform Responsibility**

Major content platforms, including social media companies, blogging platforms, and stock photo sites where large volumes of creator content are hosted, should bear a share of compliance responsibility. Platforms that enable AI scraping of their users' content without consent should be required to: notify users of AI training use, provide accessible opt-out tools at the platform level, and pass through a share of any licensing revenue to individual creators whose content was included. The RSL (Really Simple Licensing) Collective, launched in September 2025 with over 50 publishers, offers a nascent model for how platform-level coordination can extend protections to smaller players within a shared infrastructure.

## **6.3 The Freelancer Gap**

Freelance and independent creators represent a particular challenge. Research consistently shows they are among the most economically vulnerable to AI displacement and the least equipped to navigate licensing systems designed for institutional actors. This framework recommends:

- An independent creator ombudsperson role within the ATDA, specifically tasked with advocating for freelance interests in governance decisions
  - Legal aid resources for individual creators navigating disputes with AI companies
  - Periodic equity audits of compensation distribution to identify and correct systematic undercompensation of independent creators relative to institutional rights holders
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## 7. Implementation Phasing

A governance framework that cannot be implemented is not a governance framework; it is a wish list. This section sets out a realistic, phased rollout that acknowledges the practical constraints facing both regulators and AI developers, while maintaining meaningful progress toward the framework's full objectives.

The model draws on two precedents: GDPR's two-year preparation period between adoption (April 2016) and enforcement (May 2018), which gave organizations time to build compliance infrastructure; and the EU AI Act's tiered rollout by risk category, which recognizes that not all obligations can or should apply simultaneously.

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### Phase 1 - Foundation (Years 1-2): Infrastructure and Transparency

**Primary goal:** Build the technical and institutional infrastructure required for the framework to function, and establish Tier 3 transparency obligations as the universal baseline.

#### Regulatory actions:

- Establish the AI Training Data Authority (ATDA) with a mandate, staffing, and budget
- Publish binding technical standards for training data logging and documentation formats, opt-out signal protocols (building on existing robots.txt and emerging Content Signals standards), and attribution metadata embedding requirements
- Launch the AI Training Data Registry in beta, with free registration for individual creators and CMOs

#### AI developer obligations:

- Begin maintaining comprehensive training data logs for all new models (Tier 3 baseline)
- Publish model cards disclosing major training data categories and sources
- Implement technical capability to process opt-out signals; compliance with verified opt-outs is not yet mandatory but must be technically possible by end of Phase 1
- Establish internal compliance functions and designate a Training Data Compliance Officer (analogous to GDPR's Data Protection Officer requirement)

**Creator-facing actions:**

- Major content platforms required to surface opt-out tools to users
- CMOs begin accepting AI training rights registrations
- Public awareness campaign about creator rights under the framework

**What is NOT yet required:**

- Tier 1 compensation payments (infrastructure not yet ready)
- Enforcement of Tier 2 opt-outs beyond good-faith compliance
- Full audit obligations

**Precedent:** This phase mirrors GDPR's preparation period, during which organizations were expected to invest in compliance readiness before enforcement began. The EU AI Act similarly required general-purpose AI model providers to begin publishing training data summaries within 12 months of the Act's entry into force, before broader obligations applied.

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**Phase 2 - Enforcement (Years 2-4): Opt-Out Rights and Accountability**

**Primary goal:** Make Tier 2 opt-out rights legally enforceable, operationalize the Registry, and begin accountability mechanisms.

**Regulatory actions:**

- ATDA begins accepting and investigating Tier 2 opt-out violation complaints
- First enforcement actions issued for systematic non-compliance with opt-out protocols
- Registry opens fully with standardized registration for all creator categories
- ATDA publishes first annual transparency report on AI training data governance
- Binding standards issued for Tier 1 collective licensing minimum terms

**AI developer obligations:**

- Full Tier 2 compliance mandatory: verified opt-out requests must be processed within 30 days; affected works excluded from future training runs
- Tier 3 transparency database must be publicly searchable and creator-queryable
- First independent compliance audits required for large AI developers (defined by model revenue threshold)
- Pre-deployment training data disclosure filed with ATDA for all new general-purpose models



**Creator-facing actions:**

- CMOs begin negotiating first collective Tier 1 licensing pools
- Individual creators can formally register Tier 1 licensing preferences
- ATDA freelancer ombudsperson office operational

**What is NOT yet required:**

- Full Tier 1 compensation payments to individual creators (collective pools still in negotiation)
- Retroactive compliance for models trained before Phase 1 began (grandfather clause with disclosure requirement)

**Precedent:** GDPR enforcement began May 2018, two years after adoption, with the first major fines issued in 2019-2020 as regulators developed enforcement capacity. This framework similarly allows time between infrastructure establishment and active enforcement.

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**Phase 3 - Full Operation (Years 4+): Compensation and Continuous Improvement**

**Primary goal:** Activate Tier 1 compensation infrastructure, apply framework retroactively where feasible, and establish continuous improvement cycles.

**Regulatory actions:**

- First Tier 1 collective licensing pools operational; compensation distributions to registered creators begin
- Retroactivity review: ATDA assesses whether and how creators whose works were used in training prior to Phase 1 can be compensated, drawing on training data documentation collected in earlier phases
- Equity audit of Phase 1-2 compensation distribution; corrective measures issued if systematic undercompensation of independent creators identified
- Framework review cycle initiated: three-year periodic review of standards, thresholds, and compensation formulas to reflect technological and market developments

**AI developer obligations:**

- Full Tier 1 licensing payments for registered works in training datasets
- Tiered compliance costs fully operational: startup and nonprofit pathways confirmed; large commercial developers subject to full audit regime
- ShareAlike and copyleft provisions reviewed in light of Phase 1-2 technical experience

### **Creator-facing actions:**

- Individual creators begin receiving quarterly compensation statements from CMOs
- Creators can view their works' full usage history via the public attribution database
- Appeals process for compensation disputes fully operational

### **Long-term continuity:**

- ATDA mandate reviewed and renewed by legislative action every five years
  - Framework explicitly designed to adapt: new content modalities (e.g., video, 3D assets, synthetic media) incorporated through delegated regulatory updates rather than primary legislation, reducing lag between technological change and governance response
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### **A Note on Retroactivity**

The most politically and practically contested question in AI training governance is what to do about the vast volume of copyrighted work already incorporated into deployed models. Full retroactive compliance (retraining every existing model to exclude opt-out works) is likely technically infeasible at current scale and would impose enormous costs.

This framework takes a pragmatic position: retroactive retraining is not required, but retroactive compensation is a legitimate goal. AI developers should be required to document what they trained on (to the extent reconstructable), make that information available to creators and regulators, and contribute to a retroactive compensation fund proportional to their commercial revenue from models trained before the framework's adoption. The precise mechanism for this fund (whether a one-time levy, ongoing royalty, or negotiated settlement process) should be determined through the ATDA's Phase 3 review, informed by the evidence gathered in Phases 1 and 2.

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## **8. Comparison with Existing Approaches**

This framework does not emerge in a vacuum. A growing ecosystem of tools, standards, and industry initiatives has emerged since 2022 in response to creator concerns about AI training. Understanding what these approaches do and where they fall short clarifies both the need for a comprehensive regulatory framework and the technical building blocks that can be incorporated into it.

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### **8.1 DeviantArt NoAI Tags**

**What it is:** Launched in November 2022, DeviantArt introduced `<noai>` and `<noimageai>` HTML meta-tag

directives that artists can apply to their work to signal that it should not be used for AI training. All deviations on the platform are opted out by default. DeviantArt updated its Terms of Service to require third parties scraping content for AI training to respect these tags.

**Strengths:** Pioneer in creator-controlled opt-out signaling at the platform level; default-opt-out model places the burden on AI developers rather than creators; technically simple to implement; inspired adoption by ArtStation, Sketchfab, and other creator platforms.

**Limitations:** The tags rely entirely on voluntary compliance. They carry no legal force in jurisdictions where AI training is covered by copyright exceptions, and growing evidence suggests that AI companies routinely scrape content regardless. A 2023 class-action lawsuit revealed that Stable Diffusion and other foundational models had ingested over 12 million DeviantArt images despite opt-out signals. As one legal analysis noted, a web-crawler blocker is not in itself a statement of reservation of rights under EU law; it simply tells bots that access is not welcomed, which is a weaker signal than a formal rights reservation under Article 4 of the EU Copyright Directive. The tags also have no mechanism for attribution, compensation, or verification.

**Relationship to this framework:** The `noai` directive family is a practical precursor to this framework's Tier 2 opt-out protocol. This framework would give such signals legal force by requiring AI developers to treat them as binding opt-out declarations backed by enforceable penalties, closing the compliance gap that currently renders them largely advisory.

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## 8.2 C2PA Content Credentials and the Content Authenticity Initiative

**What it is:** The Coalition for Content Provenance and Authenticity (C2PA) is a formal standards body co-founded by Adobe, Microsoft, Arm, Intel, and Truepic in 2021. It produces open technical standards for embedding cryptographically signed provenance metadata ("Content Credentials") directly into digital content. The Adobe-led Content Authenticity Initiative (CAI) promotes adoption of this standard and now has over 5,000 members including Google, Meta, Amazon, TikTok, Sony, Nikon, Canon, Fujifilm, the BBC, and the Associated Press. Content Credentials act as what Adobe describes as a "nutrition label" for digital content, carrying verifiable information about who created it, when, with what tools, and whether the creator wishes to opt out of generative AI training.

**Strengths:** Technically sophisticated and genuinely interoperable. Content Credentials are cryptographically signed, tamper-evident, and persist across editing and platform transfers. Backed by an impressive and growing coalition of major technology, media, and hardware companies. In 2024, Google joined the C2PA steering committee and began integrating Content Credentials into Search; the US Department of Defense implemented it on military imagery; and Google's Pixel 10 achieved the C2PA Conformance Program's highest security tier. On track for ISO standardization. The standard explicitly supports AI training opt-out signaling within the credential metadata, which aligns directly with this framework's technical requirements for Tier 2.

**Limitations:** As of 2025, adoption among actual content on the web remains minimal. Content Credentials must be actively added by creators or their tools; the vast majority of existing digital content carries no credential metadata, and will not retroactively acquire any. The standard also has no enforcement mechanism: it

can communicate a creator's preferences but cannot compel AI companies to respect them. It was primarily designed as a provenance and authenticity tool (to combat disinformation) rather than as a rights management tool, which means compensation infrastructure is entirely outside its scope. Wikipedia's entry on the CAI notes plainly that as of 2025, "adoption is lacking, with very little internet content using C2PA."

**Relationship to this framework:** C2PA Content Credentials are the ideal technical substrate for this framework's attribution and opt-out signaling infrastructure. The Tier 2 opt-out protocol and Tier 3 attribution requirements described in this framework should be built on or interoperable with C2PA standards, leveraging existing investment rather than building from scratch. The critical additions this framework provides are legal enforceability, compensation mechanisms, and universal coverage for content that will never carry C2PA metadata.

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### 8.3 Cloudflare AI Bot Blocking and Content Signals Policy

**What it is:** In July 2025, Cloudflare launched an AI bot blocking tool allowing publishers and website operators to block all AI crawlers with a single toggle, with an optional pay-per-crawl feature enabling publishers to monetize AI bot access. In September 2025, Cloudflare launched a Content Signals Policy within its robots.txt infrastructure, giving publishers a standardized way to communicate how AI crawlers may or may not use content once accessed.

**Strengths:** Cloudflare's network position gives this tool immediate practical reach; it does not require individual creators to understand or implement technical standards. The pay-per-crawl feature begins to create a market signal about the value of content access. Rapid adoption as a default feature for all new Cloudflare customers by late 2025.

**Limitations:** Coverage is limited to websites routed through Cloudflare's network and cannot address content already incorporated into existing training datasets. The pay-per-crawl model is commercially oriented toward web publishers rather than individual creators and does not address compensation for past use. The Content Signals Policy faces the same voluntary compliance limitations as robots.txt; it communicates preferences but does not legally require AI companies to honor them.

**Relationship to this framework:** Cloudflare's tools are valuable and complementary infrastructure for this framework's Tier 2 and Tier 3 mechanisms, particularly for the platform-level opt-out signaling recommended in Section 6. However, they operate at the access layer rather than the rights layer, and their effectiveness depends on AI companies' voluntary compliance, a gap this framework's legal enforcement mechanisms are designed to close.

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### 8.4 RSL (Really Simple Licensing) Collective

**What it is:** Launched in September 2025, the RSL Collective is an AI licensing framework designed to standardize how publishers communicate what content AI systems can use and under what compensation terms.



It offers four pricing models: pay-per-crawl, pay-per-inference, subscription access, and free-with-attribution. Within its first months, over 50 publishers joined, including Ziff Davis, Yahoo, Vox Media, BuzzFeed, USA Today, and Arena Group.

**Strengths:** Addresses the compensation gap that purely technical standards (C2PA, noai tags) leave open; revenue-sharing model rather than one-time licensing fees is more aligned with ongoing AI model use; gaining meaningful publisher adoption quickly.

**Limitations:** As of early 2026, participation is entirely voluntary and concentrated among mid-to-large digital publishers. The small creator equity problem described in Section 6 applies here in full. No individual creator pathway exists. AI companies are not required to join or comply.

**Relationship to this framework:** RSL represents an important market-level precursor to Tier 1 collective licensing infrastructure. This framework would formalize and extend the RSL model by: mandating AI company participation rather than relying on voluntary uptake; extending coverage to individual creators through CMO representation; and providing regulatory oversight of compensation terms to prevent large publishers from negotiating terms that disadvantage smaller creators.

### 8.5 Summary Comparison

| Approach                      | Opt-Out                      | Attribution            | Compensation            | Legal Force | Small Creator Coverage          |
|-------------------------------|------------------------------|------------------------|-------------------------|-------------|---------------------------------|
| NoAI / <span>noai</span> tags | Yes                          | No                     | No                      | No          | Yes (platform default)          |
| C2PA Content Credentials      | Yes (signal only)            | Yes                    | No                      | No          | Partial (requires active setup) |
| Cloudflare AI Blocking        | Yes (access layer)           | No                     | Partial (pay-per-crawl) | No          | Partial (Cloudflare users only) |
| RSL Collective                | Partial (publisher focus)    | Yes                    | Yes                     | No          | No                              |
| <b>This Framework</b>         | <b>Yes (legally binding)</b> | <b>Yes (mandatory)</b> | <b>Yes (Tier 1)</b>     | <b>Yes</b>  | <b>Yes (designed in)</b>        |

The pattern across all existing approaches is consistent: they provide good technical or market-level infrastructure but lack legal enforceability, compensation mechanisms, or coverage for the long tail of individual creators. This framework is designed to provide the regulatory backbone that existing voluntary tools need to become effective.

## 9. International Scope and Jurisdictional Adaptation

This framework is designed primarily for the European Union and the European Economic Area, where the regulatory foundation (the GDPR, the EU Copyright Directive's text and data mining provisions, and the EU AI Act) provides the most developed legal infrastructure for its implementation. Readers and reviewers should understand both what that means for the framework's applicability and how its core principles might translate to other jurisdictions.

### 9.1 Why the EU Is the Right Starting Point

The EU is not an arbitrary choice of jurisdiction. Three structural factors make it the logical anchor for a rights-based AI training governance framework:

**Existing legal architecture:** The EU Copyright Directive's Article 4 already establishes the right of rights holders to opt out of commercial TDM. This framework builds on and strengthens that foundation rather than creating new legal concepts from scratch. The GDPR establishes the procedural model (consent, transparency, data minimization, right of access, enforcement with meaningful penalties) that this framework adapts to training data governance.

**Market leverage:** The EU's "Brussels Effect" (its documented tendency to set de facto global standards through market access requirements) means that AI companies wishing to operate in EU/EEA markets must comply with EU rules regardless of where they are headquartered. A framework with genuine EU legal force therefore reaches well beyond European borders. Academic analysis by Bradford (2020) and Siegmann and Anderljung (2022) has documented how EU AI regulation is already producing Brussels Effect dynamics in global AI development practices.

**Creator rights tradition:** The EU's strong moral rights tradition provides a more receptive legal and cultural environment for a framework that takes creator rights seriously, compared to jurisdictions where copyright is viewed more purely as an economic right.

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### 9.2 The Global Regulatory Landscape

Understanding how this framework relates to other jurisdictions requires a brief map of where each major AI-development region currently stands on training data copyright.

**United States:** The US has no specific legislation governing AI training and copyright. The question is being resolved through litigation, with the key legal question being whether training on copyrighted works constitutes "fair use" under Section 107 of the Copyright Act. Courts consider four factors: the purpose and character of the use, the nature of the work, the amount used, and the effect on the market for the original. As of mid-2025, courts are reaching varied conclusions, making US law highly uncertain. The US Copyright Office published a major report on AI training and copyright in 2025, but Congress has not legislated. The absence of opt-out rights, mandatory attribution, or compensation infrastructure in US law means this framework cannot be

straightforwardly applied in the US context, though voluntary adoption of its principles by US-based AI companies serving EU/EEA markets is achievable through contract and terms-of-service mechanisms.

**United Kingdom:** The UK's position is genuinely unresolved. Post-Brexit, the UK retained its existing TDM exception for non-commercial research (Section 29A, Copyright, Designs and Patents Act), but proposed a broader commercial TDM exception in 2022 and abandoned it in early 2024 after significant pushback from creators. As of August 2025, the UK government has consulted on an EU-style TDM exception with opt-out rights but has not legislated, stating it will not do so until workable opt-out mechanisms exist. The UK's current trajectory suggests it may eventually converge with the EU framework, making the infrastructure proposed here directly relevant to UK adaptation with relatively modest modifications.

**Japan:** Japan has the most AI-permissive copyright law of any major jurisdiction. Article 30-4 of the Japanese Copyright Act, amended in 2019, permits broad use of copyrighted works for AI training including for commercial purposes, as long as the use is not for "personally enjoying" the work's expression. Japan's Agency for Cultural Affairs issued clarifying guidance in May 2024 narrowing the exception somewhat, particularly for fine-tuning intended to reproduce specific works, but Japan remains a significantly more permissive environment than the EU. A framework like this one could be adopted voluntarily in Japan but would require primary legislation to have binding force.

**Singapore:** Singapore's Copyright Act Amendment of 2021 introduced Section 244, permitting use of copyrighted works for "computational data analysis," an AI-friendly exception that includes commercial use. Singapore has made no secret of its intention to be a regional AI hub. Like Japan, this framework's mandatory provisions would require legislative action to implement in Singapore.

**China:** China has broad copyright exceptions that have generally been interpreted permissively for AI training, combined with a national AI strategy that prioritizes development speed. Rights-holder leverage in China is substantially lower than in EU/EEA jurisdictions.

**Comparative summary:** A 2025 comparative legal study published in the *International Review of Intellectual Property and Competition Law* examined the US, Canada, UK, EU, Israel, China, Singapore, and Japan, concluding that "countries with broad exceptions for fair use/fair dealing, and with no specific TDM exceptions, currently provide the least guidance as to whether uses are permitted by law," while "Singaporean and EU law currently provide the most favorable conditions for AI companies" in terms of legal certainty. That certainty comes with the EU's opt-out rights attached, creating the leverage this framework requires.

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### 9.3 Jurisdictional Arbitrage and Its Limits

A legitimate concern about any EU-anchored framework is that AI companies could simply conduct training operations in more permissive jurisdictions (Japan, Singapore, or the US) and deploy the resulting models globally, thereby circumventing EU requirements.

The EU AI Act addresses this directly. Article 53(1)(c) and Recital 106 of the AI Act require providers of general-purpose AI models to comply with EU copyright law in their training activities, and the territorial reach

provisions make clear that it is "irrelevant where the acts of extraction and reproduction take place," meaning that a model trained in Japan on EU-right-holder content and then deployed in the EU must still comply with EU transparency requirements. The Brussels Effect reinforces this: AI companies that want to sell their models in the EU/EEA must comply with EU rules regardless of where training occurred.

This does not fully eliminate arbitrage risk, but it substantially limits it for commercially significant AI deployments targeting European markets, which includes virtually every major AI company currently operating globally.

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#### **9.4 Pathways for International Adaptation**

While this framework is designed for EU/EEA implementation, its core principles are exportable. The following adaptation notes apply to the major jurisdictions:

**For the US:** The Tier 3 transparency requirements and attribution database infrastructure are achievable through industry self-regulation or as conditions attached to federal AI funding, without requiring new copyright legislation. Tier 1 collective licensing could be built on existing CMO infrastructure (ASCAP, BMI, Copyright Clearance Center). Tier 2 opt-out rights would require either legislative action or reliance on contractual mechanisms through platform terms of service, the latter being weaker but immediately achievable.

**For the UK:** The UK is the most natural candidate for near-identical adoption, given its proximity to EU law and the government's own stated intention to legislate in this area. The framework could serve directly as a template for UK legislation, with minor adjustments to align with UK rather than EU regulatory structures.

**For multilateral contexts:** The World Intellectual Property Organization (WIPO) is actively considering international coordination on AI and copyright. This framework's three-tier architecture could serve as a contribution to those discussions, offering a model that is more prescriptive than purely voluntary industry codes but more practically achievable than full harmonization of copyright law across jurisdictions.

The absence of a global consensus on AI training and copyright is a genuine problem, and this framework cannot solve it alone. What it can do is demonstrate that a workable, rights-respecting governance structure is achievable in at least one major market, and provide a model that other jurisdictions can learn from, adapt, and ultimately converge toward.

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### **10. Addressing Common Objections**

**"Opt-in requirements would make AI training impossible."** Tier 1 is voluntary. It creates incentives for creators who want to participate in AI markets. Tier 2 opt-outs require affirmative action. Tier 3 imposes transparency requirements but does not restrict use. The framework is designed to be compatible with continued large-scale AI training.



**"There's no way to verify which works were used in training."** This objection is increasingly overstated. Content provenance and fingerprinting technologies are advancing rapidly, and the EU AI Act already mandates training data disclosure. The framework creates incentives for AI companies to invest in these verification systems.

**"Compensation systems would be too complex to administer."** Music royalty systems handle billions of micro-transactions annually across millions of rights holders. The infrastructure challenge is real but solved. It requires investment in collective management organizations, not invention of new concepts.

**"This would advantage large incumbents over startups."** This is a legitimate concern. The framework includes provisions for proportional compliance costs based on model revenue and scale, simplified compliance pathways for non-commercial and research AI, and grace periods for early-stage companies.

**"AI companies will just train in other jurisdictions to avoid compliance."** The EU AI Act's territorial reach provisions and the Brussels Effect substantially limit this. Article 53 makes clear that compliance obligations apply regardless of where training occurs for models deployed in the EU/EEA. See Section 9.3 for full treatment of the jurisdictional arbitrage question.

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## 11. Conclusion

The question of how AI systems should relate to the creative work they are trained on is not merely a legal technicality. It is a foundational question about what kind of AI development ecosystem we want to build. A framework that treats creator rights as an afterthought will generate ongoing legal conflict, erode public trust in AI, and ultimately harm the long-term interests of the industry itself.

The three-tier model proposed here takes a different approach: it treats creators as stakeholders with legitimate interests, designs practical mechanisms for honoring those interests, and draws on decades of proven infrastructure from adjacent industries. It is not a perfect solution. No framework could be. But it offers a workable path toward AI development that is both innovative and just.

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*This white paper was developed as a portfolio project exploring the intersection of AI ethics, intellectual property law, and regulatory policy. The framework is proposed as a contribution to ongoing public debate and does not constitute legal advice.*